

Real Racing 3 F1 Reference Gap Report

Reference reviewed: FORMULA 1 2022 Gameplay - Spa Francorchamps - Real Racing 3 Update 10.7.mp4. Purpose: compare the target race feel against the current Unity F1 project and identify what can be improved now without disrupting future phases or UI ownership.

Executive Summary

The current build has a solid Phase 2.3 foundation: shared vehicle physics, race grid spawning, racing line waypoints, AI perception, difficulty profiles, basic overtaking, and recent track recovery safeguards.

The biggest reference gap is not raw physics. It is race feel: engine and gear feedback, a clean chase camera, race-craft behavior, and presentation polish. UI should stay out of scope because Palak owns that area.

The safest immediate work is a Race Feel Pass 2.3A focused on engine sound v1, one clear chase-camera method, and AI race-craft tuning that does not replace the current AI architecture.

Reference Observations From RR3

Area	What the reference does well
Race presentation	The race feels official immediately: track intro, clear lap/position HUD, opponent labels, speed/gear display, and mobile controls.
Engine and gear feel	Constant engine pitch, clear acceleration rise, gear-change rhythm, and audible deceleration sell speed even before the camera or HUD are considered.
AI race craft	Cars use track width, curbs, entry/apex/exit lines, side-by-side spacing, and believable overtake/defend behavior instead of driving like one train.
Camera	The chase camera is close, low, stable, speed-readable, and keeps the car framed while showing enough road for braking and overtakes.
Track feel	Spa has strong edge readability: curbs, runoff, barriers, boards, elevation, and trackside landmarks.
Player feedback	The player always understands speed, lap progress, position, nearby cars, and racing context.

Current Project Gaps

Gap	Current state	Impact
Race presentation	Current scene still uses debug-style presentation. UI work is owned by Palak, so this should not be touched in this pass.	Game feels less like a final race even when physics is improving.
Engine and gears	Scene inspection found no AudioSources, no engine loop, no RPM/gear presentation model, and no engine braking layer.	This is the largest missing piece for player feel.
AI race craft	Foundation exists, but behavior is still early: lane use, defend/overtake personality, and curb/runoff decisions need tuning.	AI can feel less human and less competitive than RR3.
Camera polish	Main Camera and CinemachineBrain exist, plus custom camera target logic, but the setup is confusing and not yet a single polished method.	Speed sensation and road readability are weaker than the reference.
Track feel	Current road uses free assets. It is functional and should stay for now.	Visual authenticity is limited, but replacing the track now would distract from Phase 2.3 goals.
Player feedback	Final feedback systems should arrive step by step. No UI changes in this pass.	Needs later integration with UI and race systems.

User Clarifications Captured

- UI is out of scope for us right now. Palak owns UI, so we should not create or modify HUD, mobile controls, race presentation screens, or player feedback UI until coordination happens.
- Engine and gear feel are high priority. The right timing is to build engine sound v1 soon as part of a small Race Feel Pass, while leaving full drivetrain/engine braking for a later physics phase.
- AI race craft can be manipulated and improved now within the existing Phase 2.3 foundation. Full advanced AI remains a later phase.
- Camera should be simplified to one clear method. We should choose either Cinemachine or a custom camera path, then remove confusion from the scene setup.
- Track replacement is out of scope. Current free road assets should remain for now.
- Final player feedback is still desired, but it should arrive gradually and coordinate with UI ownership.

Safe Immediate Work Candidates

Candidate	Why it is safe now	Future compatibility
Engine sounds v1	Can use existing speed/throttle/brake data without changing drivetrain physics.	Later can be driven by real RPM, gears, and engine braking.
Cinemachine chase camera	Can polish camera feel without touching UI, track assets, or physics.	Can remain production camera or be replaced cleanly if custom camera wins.
Phase 2.3-safe AI race craft	Builds on current AIDriverController, racing line, perception, and difficulty profiles.	Can feed into Phase 3.3 adaptive AI instead of being thrown away.

Do Not Apply In This Pass

- Do not modify UI or race HUD. That area belongs to Palak and should remain coordinated separately.
- Do not replace the current track or road assets.
- Do not implement final engine simulation, final gearbox physics, or engine braking yet unless a later plan explicitly scopes it.
- Do not rewrite the AI architecture. Keep changes compatible with the current Phase 2.3 system.

Recommended Next Planning Topic

When planning resumes, define Race Feel Pass 2.3A with three isolated tracks: engine sound v1, one chase-camera method, and AI race-craft polish. The plan should explicitly exclude UI and track replacement.

Prepared for the Unity F1 project in G:/F1-Git_Repo/F1. This document is a reference report only, not an implementation patch.